

# JI YOUNG BYUN

[jbyun13@jhu.edu](mailto:jbyun13@jhu.edu) · Baltimore, MD

## PROFESSIONAL SUMMARY

Ph.D. candidate in Biomedical Engineering at Johns Hopkins University focused on developing reliable and trustworthy LLM/VLM-based AI systems for scientific reasoning and biomedical decision-making. Research and industry experience span multi-agent LLM systems for hypothesis generation, hallucination-aware calibration for medical VQA, test-time scaling for clinical reasoning, and efficient inference for vision models, with publications in COLM, MIDL, NeurIPS workshops, and Cell.

## EDUCATION

### Ph.D. Candidate in Biomedical Engineering

Aug 2022 – Dec 2026 | GPA: 3.85/4.0

#### Johns Hopkins University

- Advisor: Dr. Rama Chellappa
- Thesis: Towards Reliable Biomedical Decision-Making with Foundation Models

### M.S. in Bio and Brain Engineering

Feb 2019 – Feb 2021 | GPA: 4.0/4.0

#### Korea Advanced Institute of Science and Technology (KAIST)

- Advisor: Dr. Yong Jeong
- Thesis: Graph Neural Network for Predicting Alzheimer's Disease

### B.S. in Bio and Brain Engineering

Mar 2013 – Feb 2018 | GPA: 3.7/4.0

#### Korea Advanced Institute of Science and Technology (KAIST)

## TECHNICAL SKILLS

**Agentic AI/LLMs:** Multi-agent Orchestration, Tool-Calling, Multimodal Reasoning, Test-Time Scaling, Calibration, Prompt Engineering

**ML/AI:** Deep Learning, Computer Vision, Medical Image Analysis, Efficient Inference

**Languages:** Python, SQL, MATLAB | **Frameworks:** PyTorch, TensorFlow, HuggingFace, LangChain, Databricks

## RESEARCH EXPERIENCE

### Research Intern | Merck & Co.

Feb 2026 – Aug 2026

#### Multi-agent Framework for Toxicology Hypothesis Generation (Accepted, NeurIPS'26 AgenticLS Workshop)

- Designed and built an end-to-end multi-agent AI framework for pharmaceutical toxicology hypothesis generation, using tools to integrate enterprise-level multimodal databases for drug safety and mechanistic reasoning.
- Developed an evidence-grounding and evaluation framework to quantify LLM hypothesis faithfulness without traditional ground-truth labels, systematically comparing agent architectures.

### Graduate Research Assistant | Johns Hopkins University

Aug 2022 – Dec 2026

#### Hallucination-Aware Confidence Calibration in Medical VQA (Accepted, COLM 2026)

- Project page: <https://jiyoungbyun.github.io/vlm-hac/>
- Proposed Hallucination-Aware Calibration (HAC) in collaboration with Microsoft, integrating vision-grounded hallucination signals to improve reliability/calibration and AUROC, with the largest gains on open-ended clinical questions.
- Conducted comprehensive calibration study across general-purpose VLM families (2B–38B), 8+ confidence strategies, 3 medical VQA benchmarks; established that overconfidence is pervasive and not resolved by scaling or prompting.

#### Test-Time Scaling for Clinical Decision Making (Accepted, MIDL 2026)

- Developed a test-time scaling framework for medical decision making tasks with large reasoning models and VLMs, improving performance without costly supervised annotation across text and vision modalities.
- Achieved up to 30.4pp AUC improvement on 3 medical image diagnosis benchmarks and 17.5pp accuracy gain across 6 MMLU medical domains and derived analytical scaling laws.

#### Adaptive Inference for Medical Vision Transformers (Accepted, MIDL 2026)

- Proposed a unified adaptive inference framework for medical Vision Transformers that combines token reduction and early exiting, using JSD-based redundancy scoring and a lightweight CNN predictor for per-sample strategy selection.
- Achieved 71.4% average FLOPs reduction with only 0.1pp accuracy loss across 5 medical datasets, outperforming standalone early-exit and token-reduction strategies; demonstrated 77.2% FLOPs reduction on PathMNIST and 69.8% reduction on real-world INSIGHT cataract data while maintaining baseline accuracy.

## Smartphone-Based Cataract Detection under Data Shifts *(Accepted, NeurIPS'24 AIM-FM Workshop)*

- Built an end-to-end AI pipeline for lens status classification from smartphone-captured images, in collaboration with ophthalmologists at JHU Wilmer Eye Institute and Aravind Eye Hospital, enabling scalable cataract screening in low-resource settings and prospective clinical deployment.
- Introduced distribution probing fine-tuning strategy to address device-driven covariate shift, achieving 91% accuracy in both in-silico evaluation and real-world deployment with patients in rural India.

## Research Associate | KAIST

Feb 2019 – Apr 2022

- Led development of a GNN framework for Alzheimer's disease classification using rs-fMRI and demographic data, proposing a bipartite subject–demographic graph with APPNP, while managing a KAIST–KT Corp. \$85K/year research collaboration.

## SELECTED PUBLICATIONS

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### Auditing Pairwise Equivalence Judgments: Self-Critique Effects and Diversity Measurement in Multi-Agent Hypothesis Generation

*NeurIPS'26 AgenticLS Workshop*

Byun, J. Y., Hu, A., Rogers, J., Wang, R., Kesapragada, M., Shah, F. (2026)

### Overconfidence and Calibration in Medical VQA: Empirical Findings and Hallucination-Aware Mitigation

*COLM 2026*

Byun, J. Y., Park, Y. J., Corbeil, J.-P., & Ben Abacha, A. (2026)

### Test-Time Scaling in Clinical Decision Making: An Empirical and Analytical Investigation

*MIDL 2026*

Byun, J. Y., Park, Y. J., Azizan, N., & Chellappa, R. (2026)

### Adaptive Inference for Medical Vision Transformers: Token Reduction or Early Exit?

*MIDL 2026*

Byun, J. Y.\*, Lee, H. S.\*, Shuff, J. M., Venkatesh, R., Shekhawat, N. S., Parikh, K. S., & Chellappa, R. (2026)

### Cataract Detection from Small Datasets: Fine-Tuning and Real-World Validation under Data Shifts

*NeurIPS'24 AIM-FM Workshop*

Byun, J. Y.\*, Shuff, J. M.\*, Munoz, R., Venkatesh, R., Shekhawat, N. S., Parikh, K. S., & Chellappa, R. (2024)

### PAAN/MIF Nuclease Inhibition Prevents Neurodegeneration in Parkinson's Disease

*Cell [IF: 42.5]*

Park, H., Kam, T. I., ... Byun, J. Y., ... & Dawson, V. L. (2022)

### Graph Neural Network Based Heterogeneous Propagation Scheme for Classifying Alzheimer's Disease

*bioRxiv 2021*

Byun, J. Y. & Jeong, Y.

### Label-free Rapid Viable Enrichment of Circulating Tumor Cell by Photosensitive Polymer-based Microfilter

*Theranostics [IF: 13.3]*

Kang, Y. T., Doh, I., Byun, J. Y., Chang, H. J., & Cho, Y. H. (2017)

## HONORS & AWARDS

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Samsung Research America START Program Finalist

2025

Kwanjeong Scholarship for Overseas Study

2022 – 2026

KAIST–KT Joint Research Grant

2021 – 2022

Best Poster Award — ASMRM & ICMRI Conference

2020

National Scholarship

2019 – 2021

National Science & Technology Scholarship

2013 – 2017

## MENTORSHIP & TEACHING

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### Teaching Assistant | *Deep Learning for Medical Imaging*

Jan 2025 – May 2025

- Designed and delivered weekly coding sessions on medical image analysis with PyTorch; graded quizzes and held office hours for 20+ graduate students.

### Research Mentor | *Johns Hopkins University*

Mar 2023 – Present

- HyeonSeo Lee (JHU BME): Mentoring on token reduction strategies for Vision Transformers in medical imaging.
- Lisa Hou (JHU BME): Guided fine-tuning and evaluation of segmentation models on real-world ophthalmology images.